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Data Science Fundamentals

Polynomial Regression

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Before we get started…

Practicalities

Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory excercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

Syllabus

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression	Week - 11	Clustering - Hierarchical clustering
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5	Classification - Decision trees	Week - 13	Regularization: Lasso regression
Week - 6	Classification - kNN	Week - 14	Recap: Theory + Excercises
Week - 7	Clustering - kMeans	Week - 15	Exam Preperation + Q&A
Week - 8	Lifesnaps guidance		

(subject to change)

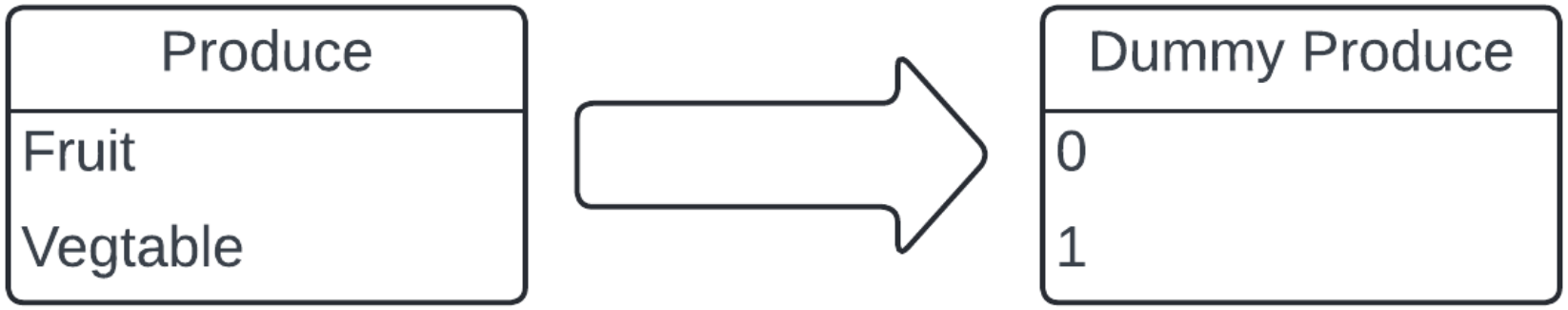
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Now we get to the fun part

Polynomial Regression

Dummy variables

A dummy variable is a **numeric variable** that represents **categorical data**, such as gender, race, political affiliation etc... It only takes the value 0 or 1 to indicate the absence or presence of said effect that may be expected to shift the outcome.

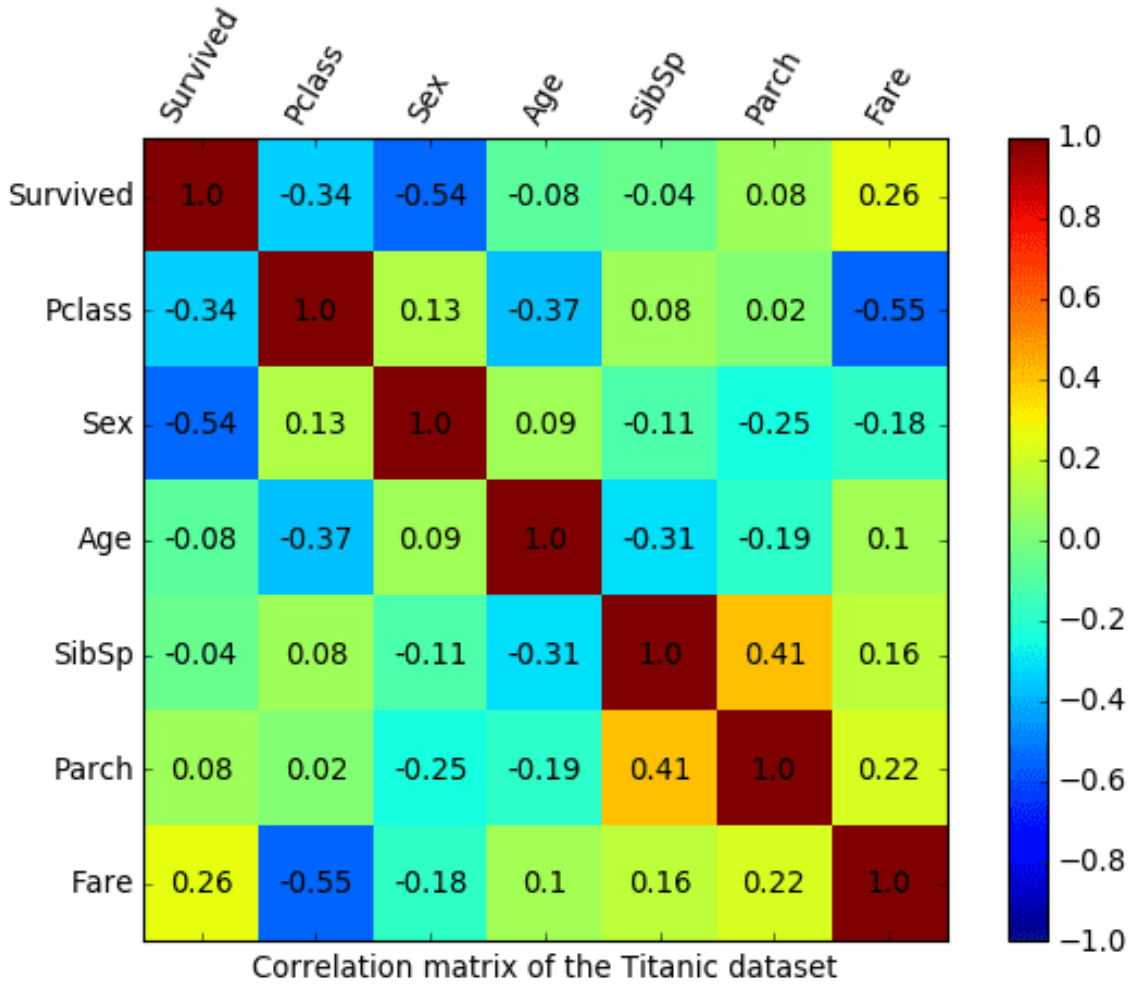


Interaction effect

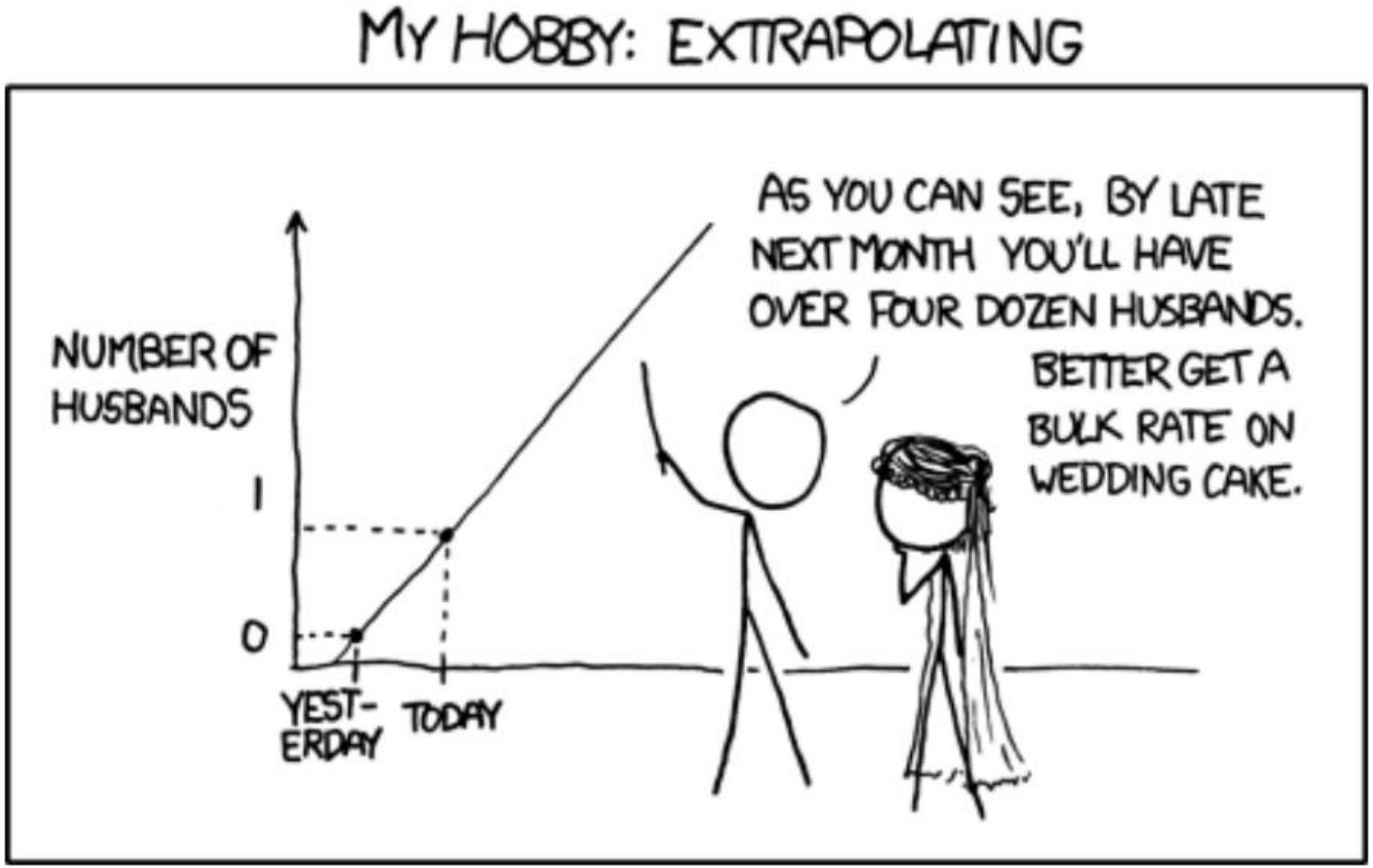
- Occurs when the effect of one variable depends on the value of another.
- Captures **combined influence** of variables, beyond their individual effects.
- Often represented by multiplying two variables (e.g., ‘Years of education’ x ‘Seniority’).
- Useful for modeling more complex relationships in data.
- Helps reveal if variables together have a stronger or different impact than separately.

Correlation matrix

- Also known as a “heatmap”
- Shows correlation between all possible variable pairs
- Displays in a matrix form
- “How well do two variables explain each other?”
- Central axis is always “1”
- Both sides are mirrored images of each other

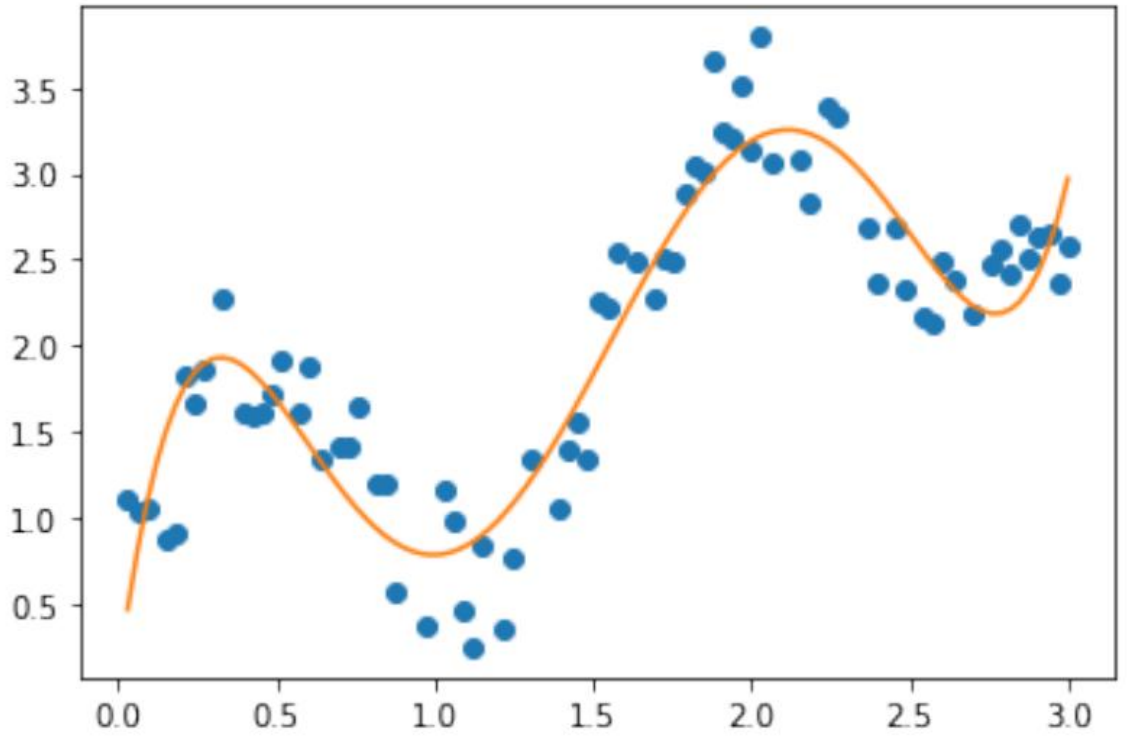


Polynomial regression



Polynomial regression

- The ‘upgraded’ form of regression
- Linear regression is cool, but limited.
- Your model **moves** with the data
- “line of best fit”



Polynomial relation = polynomial function

- We can describe this behavior as a mathematical function

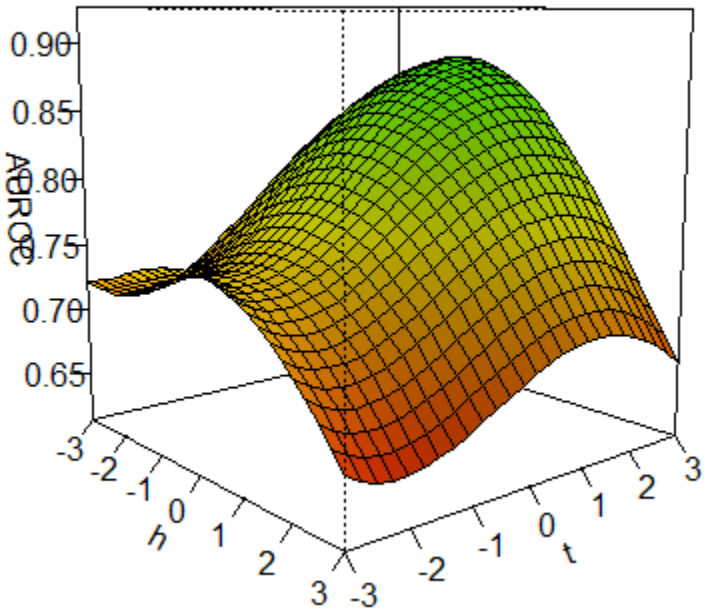
$$f(x) = ax + bx^2 + cx^3 + dx^4 + \ldots + e$$

$$y_i = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4 + \ldots$$



Multiple polynomial regression

- Same as polynomial regression, but with **multiple factors**
- Example: “# of ice creams sold” is affected by “temperature” and “hours of sunlight”
- “plane of best fit”
- Our function becomes increasingly complex…
  - → Luckily python handles this for us!



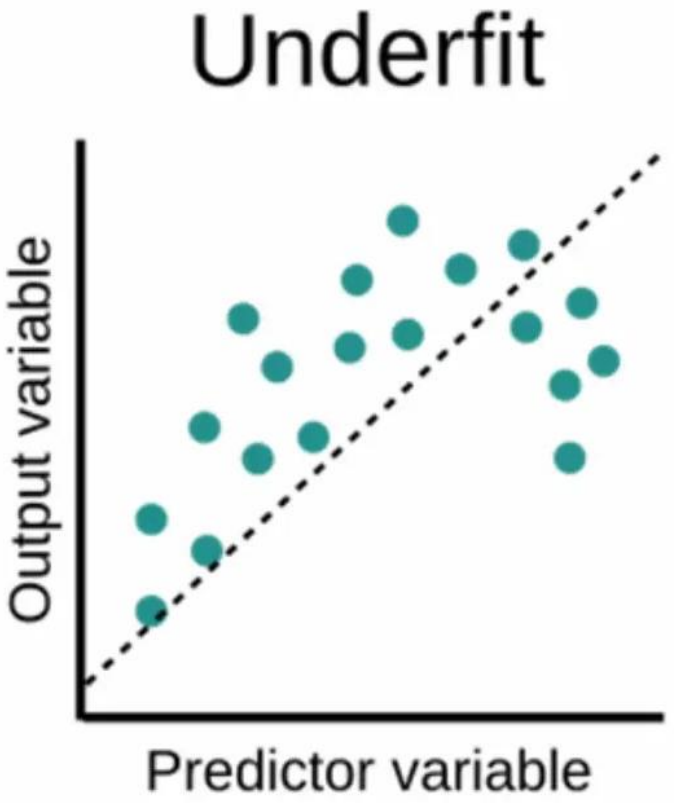
Evaluation

- Since it is still regression the same metrics apply
  - R^2
  - MAE
  - MSE



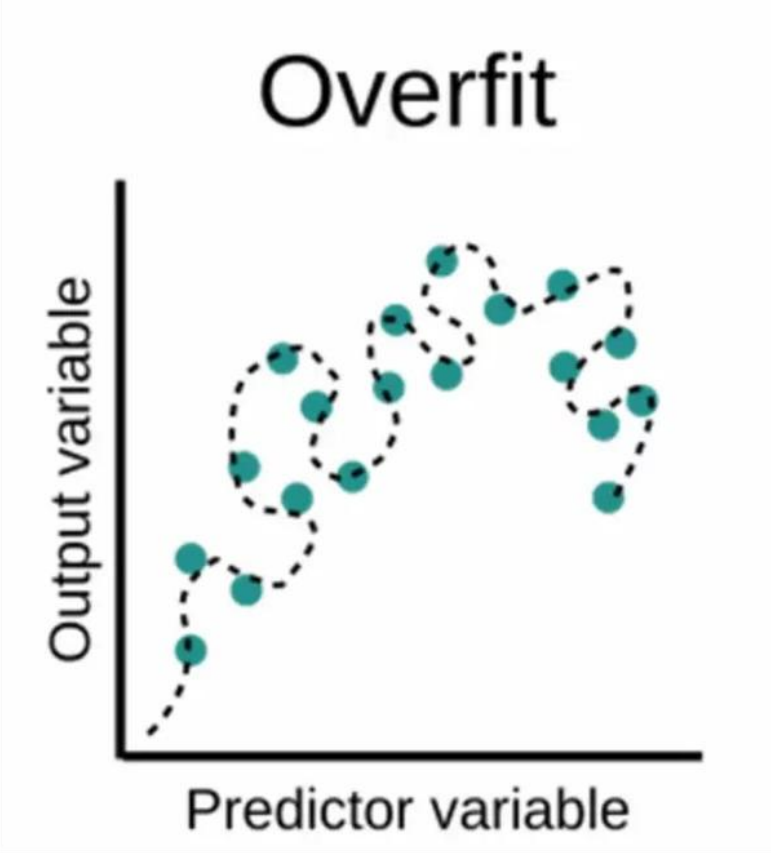
Underfitting

- Model is too simple to capture underlying patterns.
- Model **fails to learn** from the training data.
- Poor performance on both training and test data.

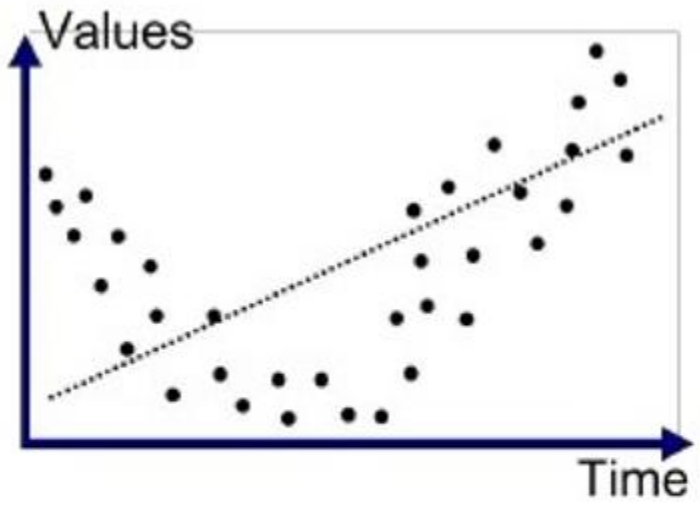


Overfitting

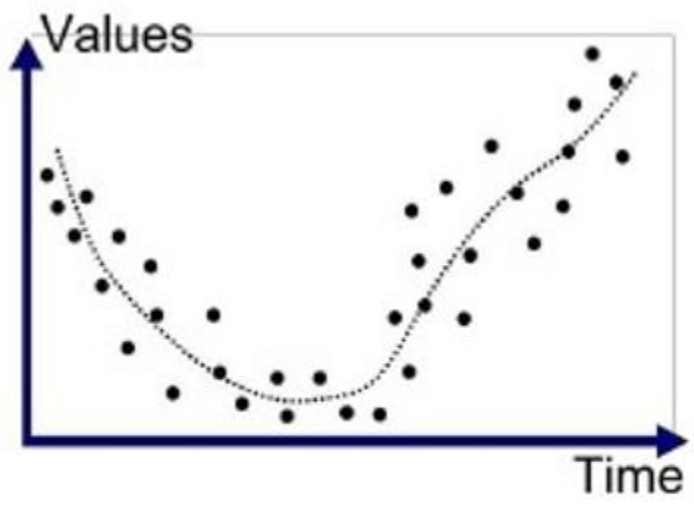
- Model is too complex, capturing noise and random fluctuations.
- Model **learned too well** from training data
- Performs well on training data but poorly on new, unseen data.



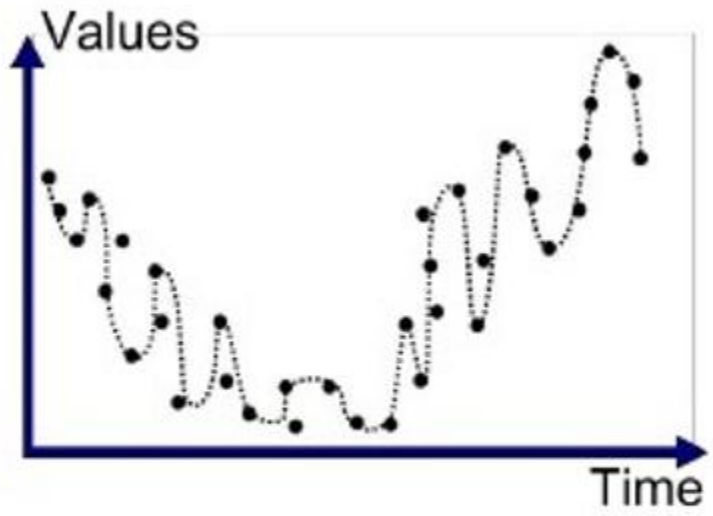
Finding the right balance



Underfitted



Good Fit/Robust



Overfitted

Source: Anup Bhande, What is underfitting and overfitting in machine learning and how to deal with it. (Medium 2018)

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Let's jump into action

Excercises

First, let's explore this together

- Go to canvas and download the notebook "DSF 03 - Polynomial Regression Notebook"
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Follow along!



Now it's up to you!

- Go to canvas and download the notebook "DSF 03 - Polynomial Regression Exercise"
- Also download the csv file called "Auto.csv"
- Place them in the same directory
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Go through the notebook and make the excercises

I don't bite, ask me any questions you have!

